

KRAKENES FYR, Norway

WMO: 012030

Lat: 62.0342N

Lon: 4.9862E

Elev: 41

StdP: 100.83

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.9	-1.7	-15.4	1.0	6.4	-12.5	1.3	2.3	26.6	6.4	24.7	6.1	5.0	80	0.911

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.3	21.3	15.3	19.6	14.6	18.2	14.1	16.8	20.0	15.9	18.5	15.1	17.5	5.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	11.2	17.4	14.7	10.5	17.0	13.8	9.9	16.3	47.3	20.3	44.7	18.6	42.4	17.5	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 25.1	22.4	20.1		-3.6	24.6	1.7	2.5	-4.8	26.3	-5.8	27.8	-6.7	29.2	-7.9	30.9	
(5)			DB	-5.7	18.1	1.9	1.1	-7.1	18.9	-8.2	19.6	-9.2	20.2	-10.6	21.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	3.6	3.4	3.9	6.3	8.7	11.2	14.0	14.5	12.8	9.2	6.3	4.6
	DBStd	4.77	2.41	2.47	2.40	2.80	3.09	2.39	2.76	2.32	2.47	2.85	2.91	2.84
	HDD10.0	1101	197	184	190	117	63	12	1	1	4	48	115	168
	HDD18.3	3695	456	417	448	362	300	214	139	121	166	283	362	427
	CDD10.0	454	0	0	0	5	22	48	124	141	88	23	3	0
	CDD18.3	9	0	0	0	0	0	1	4	3	1	0	0	0
	CDH23.3	13	0	0	0	0	0	0	11	2	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
(14) Wind	WSAvg	8.6	9.8	9.3	9.0	8.5	7.6	7.4	6.7	7.0	9.0	9.4	9.4	10.3
(15) Precipitation	PrecAvg	1857	203	170	158	110	83	98	111	141	165	197	198	223
	PrecMax	2451	364	342	321	214	130	220	174	253	289	357	367	398
	PrecMin	1315	37	16	34	28	20	12	26	55	56	61	46	55
	PrecStd	286	86	78	69	38	28	45	36	46	73	69	84	88
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	10.7	9.9	16.2	20.3	21.1	24.6	22.8	21.1	19.0	14.2	11.5
		MCWB	4.3	6.5	6.4	8.6	12.6	12.9	16.6	17.0	14.9	13.3	9.6	7.3
	2%	DB	8.2	8.4	8.5	13.7	17.5	18.1	21.7	20.7	19.1	15.7	12.0	9.9
		MCWB	6.3	5.8	5.6	8.2	11.6	12.3	16.2	14.9	13.6	10.3	8.6	7.3
	5%	DB	7.5	7.5	7.7	11.9	14.9	15.7	19.6	19.1	17.5	13.9	11.0	9.0
		MCWB	5.7	5.3	5.5	7.4	10.3	11.8	15.4	14.9	13.0	10.4	8.2	7.2
	10%	DB	6.9	6.8	6.9	10.2	12.8	14.1	17.9	17.6	16.2	12.8	10.1	8.3
		MCWB	5.1	4.9	4.9	6.6	9.7	11.1	14.9	14.5	12.7	9.8	7.7	6.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	7.9	7.9	10.9	15.1	14.7	18.5	17.9	16.1	13.4	10.6	9.0
		MCDB	8.2	9.0	8.4	13.4	18.8	18.4	20.7	20.0	18.9	16.4	12.4	10.0
	2%	WB	7.1	6.9	6.9	9.3	12.3	13.6	17.2	16.6	14.9	12.1	9.8	8.2
		MCDB	7.7	7.8	7.8	12.2	15.8	16.8	20.6	19.1	17.7	14.3	11.2	9.2
	5%	WB	6.3	6.1	6.2	7.7	11.0	12.7	16.3	15.8	14.0	11.3	8.8	7.5
		MCDB	7.3	7.1	7.3	10.9	14.1	15.0	19.6	17.8	16.3	13.4	10.3	8.5
	10%	WB	5.3	5.2	5.3	6.9	9.8	11.8	15.3	15.1	13.2	10.3	7.8	6.5
		MCDB	6.6	6.5	6.6	9.7	12.7	13.5	18.2	17.0	15.3	12.3	9.7	7.8
(35) Mean Daily Temperature Range	5% DB	MDBR	2.9	2.8	3.1	3.6	3.7	3.4	3.5	3.3	3.3	3.2	3.0	2.8
		MCDBR	3.4	3.5	4.0	6.1	6.8	6.0	6.4	5.3	4.9	4.6	3.8	3.6
	5% WB	MCWBR	3.0	3.0	3.1	2.9	3.2	3.3	3.1	2.9	2.8	3.3	2.9	3.4
		MCWBR	3.0	3.4	3.5	5.4	6.6	5.2	6.0	4.3	4.2	4.5	3.6	3.6
(40) Clear-Sky Solar Irradiance	taub	0.288	0.329	0.349	0.372	0.362	0.358	0.371	0.365	0.353	0.328	0.289	0.266	0.266
		taud	2.083	2.223	2.301	2.320	2.416	2.466	2.455	2.470	2.471	2.372	2.151	2.385
	Ebn at Noon	368	616	754	813	857	868	844	818	755	622	368	208	208
		Edh at Noon	37	66	86	103	101	98	97	88	74	56	34	23
(44) All-Sky Solar Radiation	RadAvg	0.16	0.62	1.68	3.36	4.84	5.09	4.59	3.47	2.04	0.94	0.26	0.10	0.10
	RadStd	0.03	0.11	0.27	0.36	0.33	0.57	0.38	0.37	0.29	0.15	0.05	0.01	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	+0.68	N/A	N/A	N/A	N/A	-92	-135	N/A

Nomenclature: See separate page